TECHGARDEN
FRANKFURT

TECHGARDEN #1 · NOVEMBER 12, 2026

Comparing AI models.

Open and closed. European and non-European.
What can they really do?

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Hosted by HANF Advanced Fire Engineering

TECHGARDEN
FRANKFURT

WELCOME TO TECHGARDEN #1

Tonight we build together.

6 pm	Then	~7:30 pm	~8:30 pm	9 pm
Welcome and a short talk	Build in small groups	Pizza	Closing round	End

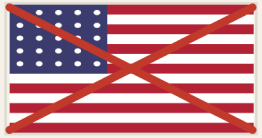


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Pizza sponsored by **TechGarden Frankfurt**

THE HOOK

A post went viral on LinkedIn

~~Anthropic~~ ~~OpenAI~~  →  **Open models from China**

“Revolut banned OpenAI and Anthropic. For over 2,000 developers.”

“The replacement: self-hosted open-weight models from China.”

“AI costs down 90 percent. Teams shipped 70 percent faster.”

“No API ping to California.”

Claims paraphrased from widely shared LinkedIn posts, November 2026

THE SOURCE

What the talk actually said

The post says	The talk says
“Revolut banned OpenAI and Anthropic.”	“We keep using Codex, Claude Code … but we keep them as exception.” Open models are the default, not the only option.
“Self-hosted … no API ping to California.”	Weights run “on a US-based service.” The US counts as an acceptable location for them.
“AI costs down 90 percent.”	The bill is “one-tenth” of before. But that leaves out the hardware (56 × B200, about $2M if bought), staff and monitoring.
“Teams shipped 70 percent faster.”	Faster, yes. But limits, runtime and harness changed at the same time. The talk doesn’t say how much came from the model.

Source: David Abitbol (Revolut), “Shipping like a startup, regulated like a bank,” AI Engineer NYC, October 13, 2026

Lots of noise.

Tonight we check for ourselves.

THE BASICS

Three things to know

What is a language model?

A program trained on huge amounts of text. It reads text and writes text.

Closed or open

Closed: you use it through the provider. **Open:** you download the weights and run it yourself.

How to try one

In a browser chat, through an API, or locally with Ollama or LM Studio.

KEY TERMS

Six words you'll hear tonight

Quantization · compressing a model, like a JPEG photo: e.g. 16 GB shrinks to 5 GB, slightly less precise.

Context window · how much text a model can take into account at once.

Hallucination · a confident answer that is made up.

Token · the unit a model reads and writes, roughly ¾ of a word. Prices and limits count tokens.

Agent · a model that plans and uses tools over several steps to reach a goal.

MCP · Model Context Protocol, a standard way to connect models to tools and data.

A WORD ABOUT THE WORD

Harness or scaffolding?



THE USUAL WORD

Harness

A horse harness steers raw power with reins. But nobody pulls a model by the reins.



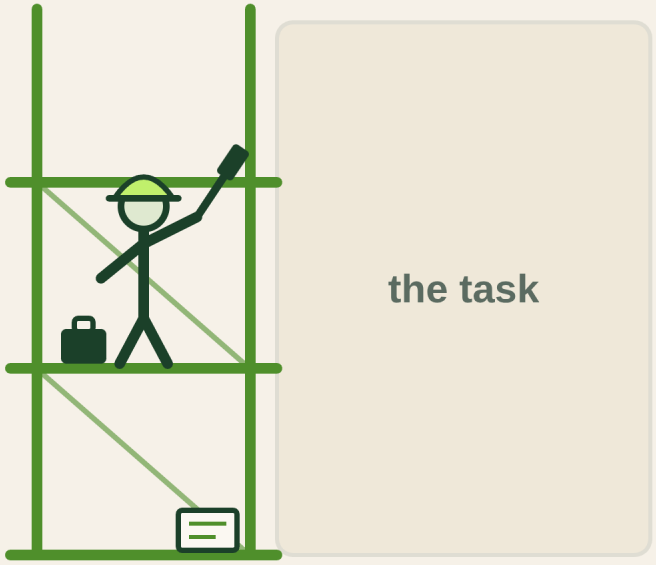
THE BETTER PICTURE

Scaffolding

A structure the model works on, like a worker: tools, memory, instructions, steps.

Tonight we say **harness**, because everyone does. **Think scaffolding.**

The worker and the scaffolding



MODEL · THE WORKER, OR THE HORSE

Does the work. Skill and knowledge are in the model.

Claude · GPT · Gemini · Mistral · Qwen · Llama

HARNESS · SCAFFOLDING AND TOOLS

Decides what the worker can reach: files, tools, plans, the next step.

Chat · Claude Code · Codex · Qwen Code · Cursor · Copilot

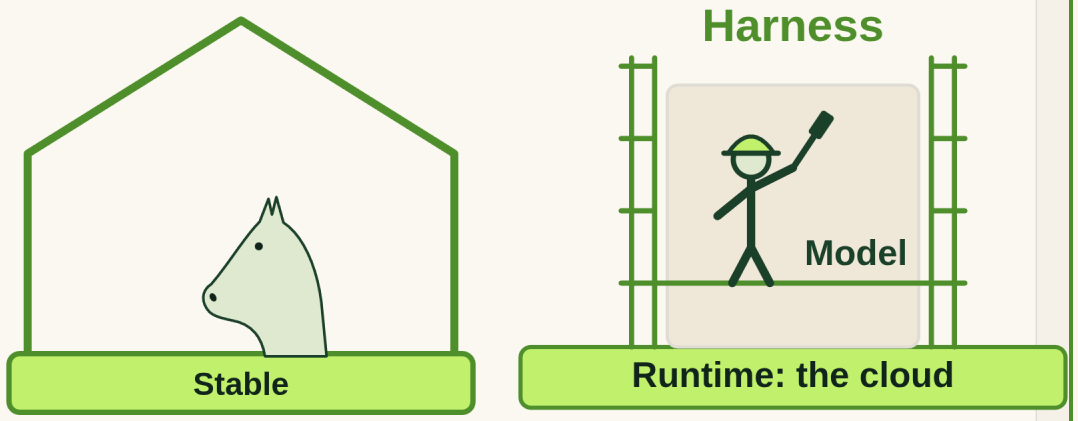
Same worker, different scaffolding, different result. Always note both.

For a fair model comparison: put every worker on the same scaffolding, e.g. the OpenRouter chat.

THE OVERLOOKED LAYER

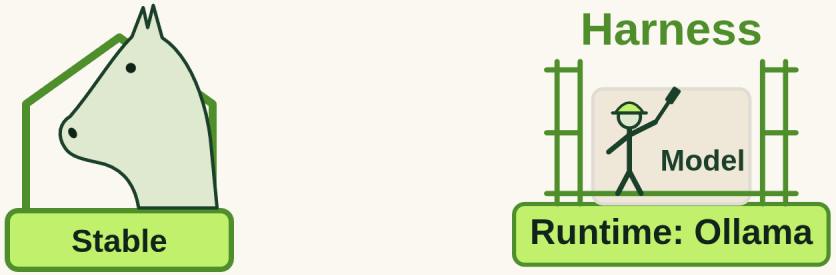
Room to move: the runtime

CLAUDE CODE · CODEX



Harness on your laptop, the rest in the cloud. **Set up for you.**

QWEN CODE + OLLAMA



Everything on your laptop. **Context, memory, speed: up to you.**

Tested yesterday: Ollama’s default context of 4,096 tokens cut off Qwen Code’s instructions, and the model looked dumb. With 16,384 it worked, using 7.9 GB of RAM.

What a bigger stable costs

The bigger the model, the more memory it needs. A small model fits on a laptop. A model good enough for real work needs a machine with a lot of memory.

What you buy	Runs	Price, approx.
A normal laptop (like mine)	small models, slowly	already there
Mini PC or Mac with 128 GB of shared memory	good models	from $2,400
NVIDIA DGX Spark, a desktop AI box	good models	$6,950
Gaming PC with a top graphics card	mid-size models, fast	card alone over €5,000
A server room, like Revolut (56 × NVIDIA B200)	the biggest models	millions

Plus the hidden cost: someone has to tune model, harness and runtime to each other, and keep them running. That’s people’s time, not hardware.

Prices from vendor and press reports, March to November 2026. Memory shortages keep pushing them up.

SOVEREIGNTY

Sovereignty has layers

Data	Where is it processed and stored?
Operation	Who runs the model, on whose servers?
Model	Who trained it, and who ships the next version?
Hardware	Whose chips does it run on?

Revolut: own operation and an open model, but US hosting and NVIDIA chips.

Local is not the same as trusted: the Bundeswehr is phasing out Chinese drones and cars, US agencies banned DeepSeek. A local model can’t phone home, but nobody can fully check what it was trained to do.

SOVEREIGNTY HAS A PRICE

Europe pays with quality

Artificial Analysis Intelligence Index: the same set of tests for every model, score 0–100



Mistral Large 4 is the best open model from the West, but 20 points behind the top and behind China’s open models.

Source: Artificial Analysis Intelligence Index v4.3.2, October 6, 2026 (Mistral Large 4 preview), via OfficeChai

TONIGHT

Same task, several models

1. Pick a task
2. Run it on two or more models
3. Note the results in results.csv
4. Open a pull request

1 · INDUSTRY TASKS · HANF ADVANCED FIRE ENGINEERING GMBH

Analyze a BIM model · Automated model checking with IDS · Sort project emails

Start here if you can. Mr. Hanf, the founder, is here tonight to answer questions. Strong solutions may lead to a follow-up.

2 · General tasks: summarize a German technical text · read a table from a PDF · say “I don’t know” · write code

github.com/techgarden-frankfurt/ai-model-comparison

Local AI or AI-built software?

WAY A

AI runs locally

Real data → **Local model** → Result

A model on your own hardware reads the confidential data. Nothing leaves the house.

Fits fuzzy work: free text, emails.
Costs: hardware, lower quality, results vary.

WAY B

AI builds, software runs

Sample data → **AI writes code** → Program

Real data → **Program** → Result

Fits rules: IDS checks, BIM analysis.
Gains: no AI sees the real data, same result every time.

Tonight: IDS and BIM → Way B · Emails → Way A, or rules plus a local model.

EXAMPLE RESULT · TASK I1

My test yesterday: one IFC model, six setups

Model · harness	Data	Result
In a chat · Python in the provider’s sandbox		
Qwen Max 4 Preview · chat	CN	✓ Counts match, found doubled stairs
Claude Opus 5.5 · chat	US	✓ Counts match, most extra findings
GPT-6.1 Sol · chat	US	✓ Counts match, few extras
Agent writes a script · Way B		
GPT-6 · Codex	US	✓ Ran on the first try
Claude Opus 5.5 · Claude Code	US	✓ Ran after one fix
Agent on my laptop · local		
qwen3:8b · Qwen Code + Ollama	local	✗ Nothing after 50 minutes

All five that worked agree: 10 storeys · 49 doors · 57 windows · no rooms, so no floor area. Not yet checked against a reference.

Run the scripts yourself, no AI needed: pip install ifcopenshell · python claude-code.py model.ifc

Model: “Einfamilienhaus” from the Bundesingenieurkammer. Details in the repo: results/2026-11-11_narges_task-I1.md

RECOMMENDED SETUPS

Which model, which harness

1 · INDUSTRY TASKS

Solve the task. Respect the data rules.

Cloud: Claude Code or Codex

Data stays on the laptop: Ollama with a small model, e.g. qwen3:8b

Public sample models and made-up emails only. Nothing confidential into any model. Note where the data went: US, EU or local.

2 · GENERAL TASKS

Same prompt, several setups.

MODEL	HARNESS	TYPE · ORIGIN
Claude	Claude Code or chat	closed · US
GPT	Codex or ChatGPT	closed · US
Qwen	Qwen Code	open · CN
Mistral	Le Chat	closed · EU
qwen3:8b	Qwen Code + Ollama	open · laptop

Good to know: open weights don’t mean a free service. Qwen Code in the cloud sends your data to Alibaba, a Chinese provider: check the region. Chat rejects .ifc? Rename a copy to .txt.

GROUND RULES

Six rules for the evening

- ✓ Public data only
- ✓ Note model, version, harness, date
- ✓ Nobody has to be a pro
- ✓ No API keys in the repository
- ✓ Bad results count too
- ✓ Ask when you’re stuck

FOR THE CLOSING ROUND

Will everything run locally in three years?

More and more, yes

Laptop models today are about as good as the frontier one or two years ago.

Models get smaller and more efficient, laptops get more and faster memory.

Everyday tasks will often stay on the device.

Not everything

Frontier models like Opus or Fable are not released as weights, and far too big.

The frontier keeps moving, and training always needs data centers.

Cheaper AI means more use: more agents, longer tasks, video.

Open question: are we building too many data centers? Some compare it to the fiber boom around 2000.

TECHGARDEN
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Come and build something.

Every other Thursday, 6–9 pm, in Frankfurt. Bring your laptop and a project.

Join the next meetup ↗ **techgarden-frankfurt.de ↗**



Thanks to our sponsor for hosting tonight
Hanf Advanced Fire Engineering

Pizza sponsored by **TechGarden Frankfurt**

BUILD. CODE. SHIP.

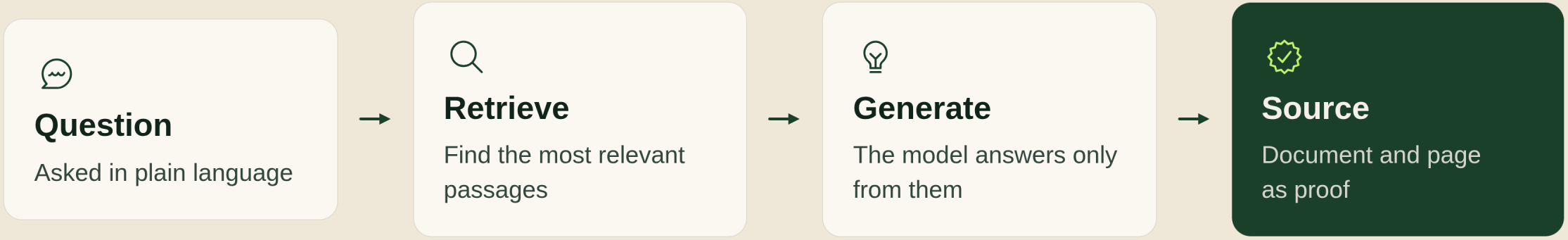
BACKUP

For questions.

RAG and NVIDIA.

RAG · RETRIEVAL-AUGMENTED GENERATION

Answers from your own documents



Why it matters: company knowledge without retraining the model, and answers you can check.

Watch out: if the search misses the right passage, the answer misses it too.

THE BOTTOM LAYER

Why almost everything runs on NVIDIA

Hardware fits the job

Thousands of cores, Tensor Cores for matrix math, very fast memory.

CUDA since 2007

PyTorch, vLLM and FlashAttention land there first. A network effect.

Scaling out

NVLink and data-center networking make many GPUs act as one.

Catching up: AMD ROCm · Apple MLX · Google TPU